Single-lead AI-ECG across three public cohorts

ECG-derived HYP labels | Simulated noise | Retrospective evaluation

DEVELOPMENT

PTB-XL
21,799 records
Lead I · 100 Hz · 10 s

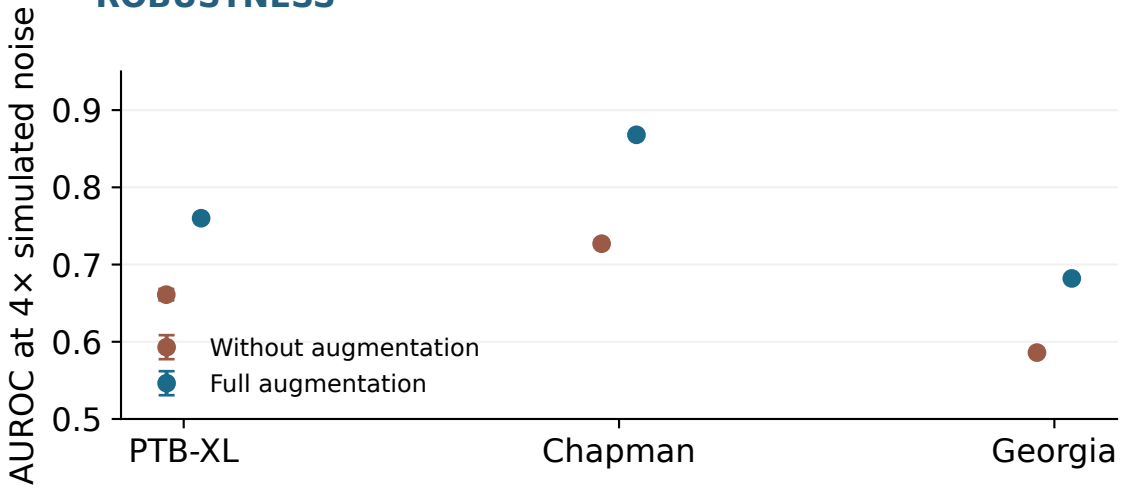
MODEL

52,225-parameter CNN
Baseline comparisons
Component-level ablation

EXTERNAL VALIDATION

Chapman: n = 2,769
Georgia: n = 10,344
Frozen model weights

ROBUSTNESS



HELD-OUT RECALIBRATION

20% calibration → 80% evaluation

Mean ECE across 10 stratified splits

Chapman 0.100 → 0.029

Georgia 0.274 → 0.006

Baseline wander carried most of the robustness benefit.

No echo-confirmed or real-wearable validation. Chapman calibration reflects case-control prevalence.